

Lid-Driven Cavity Flow: A Finite-Volume Implementation

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1 Problem Description

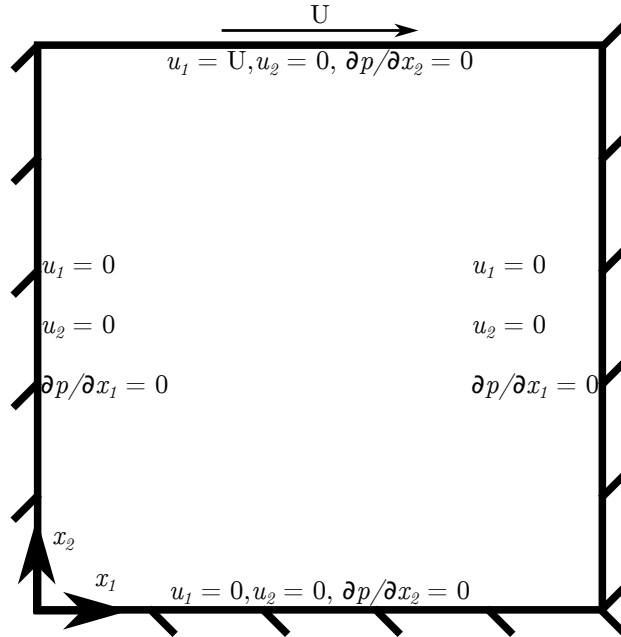


Figure 1: Lid-driven cavity flow problem illustration.

2 Governing Equations

2.1 Differential Equations

The form of the incompressible Navier-Stokes equations, and their subsequent non-dimensionalization are taken from Ferziger and Perić [1]. Dimensional quan-

tities are shown with a “*” superscript, and non-dimensional quantities are shown without modification. Conservation of mass, with dimensional units is shown in Eq. 1. Similarly, the dimensional form of the momentum equation is given in Eq. 2.

$$\frac{\partial u_i^*}{\partial x_i^*} = 0 \quad (1)$$

$$\frac{\partial u_i^*}{\partial t^*} + \frac{\partial(u_j^* u_i^*)}{\partial x_j^*} = \nu^* \frac{\partial^2 u_i^*}{\partial x_j^2} - \frac{1}{\rho^*} \frac{\partial p^*}{\partial x_i^*} \quad (2)$$

The dimensional quantities of Eq. 1, 2, are nondimensionalized according to Eqs. 3 - 6.

$$t = \frac{t^*}{t_0^*} \quad (3)$$

$$x_i = \frac{x_i^*}{L_0^*} \quad (4)$$

$$u_i^* = \frac{u_i^*}{v_0^*} \quad (5)$$

$$p = \frac{p^*}{\rho^* v_0^{*2}} \quad (6)$$

Accordingly, the nondimensional form of the continuity and momentum equation are shown in Eq. 7 and 8, respectively. In these equations, L_0^* is the reference length dimension, t_0^* is the reference time dimension, v_0^* is the reference velocity dimension, and ρ^* is the density of the fluid.

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (7)$$

$$\boxed{\text{St} \frac{\partial u_i}{\partial t} + \frac{\partial(u_j u_i)}{\partial x_j} = \frac{1}{\text{Re}} \frac{\partial^2 u_i}{\partial x_j^2} - \frac{\partial p}{\partial x_i}} \quad (8)$$

As a result of the nondimensionalization, the dimensionless numbers, Re (Reynolds number) and St (Strouhal number) appear in Eq. 8. While Eqs. 7 and 8 mathematically express mass and momentum conservation, respectively, with regards to implementing these numerically, the continuity equation (Eq. 7) does not provide any additional information that is not contained in the momentum equation (Eq. 8). The continuity equation must be further manipulated into a form which can be implemented to enforce mass continuity. Computing the divergence of the momentum equation (Eq. 8) yields Eq. 9. Simplifying Eq. 9 produces equation Eq. 10. Applying the continuity equation to Eq. 10 yields, Eq. 11, the pressure Poisson equation. In solving the pressure Poisson equation, mass conservation is enforced. Note that there is not a fluid equation

of state for closure in which *absolute* pressure appears as a term. In incompressible flow, only *gradients* of pressure appear in the momentum equation, thus by solving Eq. 11, a pressure field which satisfies continuity is determined.

$$\frac{\partial}{\partial x_i} \left(\text{St} \frac{\partial u_i}{\partial t} \right) + \frac{\partial}{\partial x_i} \left(\frac{\partial(u_j u_i)}{\partial x_j} \right) = \frac{\partial}{\partial x_i} \left(\frac{1}{\text{Re}} \frac{\partial^2 u_i}{\partial x_j^2} \right) - \frac{\partial}{\partial x_i} \left(\frac{\partial p}{\partial x_i} \right) \quad (9)$$

$$\text{St} \frac{\partial}{\partial t} \left(\frac{\partial u_i}{\partial x_i} \right) + \frac{\partial}{\partial x_i} \left(\frac{\partial(u_j u_i)}{\partial x_j} \right) = \frac{1}{\text{Re}} \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_j} \left(\frac{\partial u_i}{\partial x_i} \right) \right) - \frac{\partial}{\partial x_i} \left(\frac{\partial p}{\partial x_i} \right) \quad (10)$$

$$\boxed{\frac{\partial}{\partial x_i} \left(\frac{\partial(u_j u_i)}{\partial x_j} \right) = - \frac{\partial}{\partial x_i} \left(\frac{\partial p}{\partial x_i} \right)} \quad (11)$$

2.2 Integral Equations

The integral forms of the non-dimensional momentum equation (Eq. 8) is required to apply a finite volume method numerical solution. The integral form of the non-dimensional momentum equation is obtained by integrating the differential form of the equation over a volume domain, Ω , which yields Eq. 12. Gauss's Divergence Theorem is applied to Eq. 12 to transform the volume integrals over a volume domain Ω into surface integrals over a surface boundary σ . Furthermore, the constant non-dimensional terms are moved outside of the integrals, producing Eq. 13. This integral form of the nondimensional pressure equation, Eq. 13 is used for solving control volume problem, and thus is intuitive for applying to the uniform two-dimensional cells of this problem. The non-dimensional pressure Poisson equation, Eq. 11, is left in differential form.

$$\int_{\Omega} \text{St} \frac{\partial u_i}{\partial t} d\Omega + \int_{\Omega} \frac{\partial(u_j u_i)}{\partial x_j} d\Omega = \int_{\Omega} \frac{1}{\text{Re}} \frac{\partial^2 u_i}{\partial x_j^2} d\Omega - \int_{\Omega} \frac{\partial p}{\partial x_i} d\Omega \quad (12)$$

$$\boxed{\text{St} \int_{\Omega} \frac{\partial u_i}{\partial t} d\Omega + \int_{\sigma} u_j u_i n_j d\sigma = \frac{1}{\text{Re}} \int_{\sigma} \frac{\partial u_i}{\partial x_j} n_j d\sigma - \int_{\sigma} p n_j d\sigma} \quad (13)$$

3 Discretization

3.1 Domain and Grid

The domain of the problem depicted in Fig. 1 is described in dimensional and non-dimensional quantities in Eqs. 14, 15, respectively. A staggered grid is implemented, such that pressure nodes are placed in the center of the cells ($x_1 = x_{1,M}$, $x_2 = x_{2,M}$), u_1 nodes are placed at the mid-height of the west and east faces of the cells ($x_1 = x_1$, $x_2 = x_{2,M}$), and u_2 nodes are placed at the mid-width of the south and north faces of the cells ($x_1 = x_{1,M}$, $x_2 = x_2$), where the subscript “ M ” indicates “mid-point”. The grid is uniform, as described in

Table 1: Coordinate locations for different node types.

Node type	x_1	x_2
u_1	$m\Delta x$	$n\frac{\Delta x}{2}$
u_2	$m\frac{\Delta x}{2}$	$n\Delta x$
p	$m\frac{\Delta x}{2}$	$n\frac{\Delta x}{2}$

Eq. 16. An index “ m ” is used for the x_1 coordinate, and an index “ n ” is used for the x_2 coordinate. The coordinates of the the u_1 , u_2 , and pressure (p) nodes are tabulated in Table 3.1, for clarity, and the grid is illustrated in Fig. 2. There are ghost-nodes outside of the boundary, in both the x_1 and x_2 directions to enforce boundary conditions.

$$x_1^* \in [0, L_0^*] x_2^* \in [0, L_0^*] \quad (14)$$

$$x_1 \in [0, 1] x_2 \in [0, 1] \quad (15)$$

$$\Delta x \equiv \Delta x_1 = \Delta x_2 \quad (16)$$

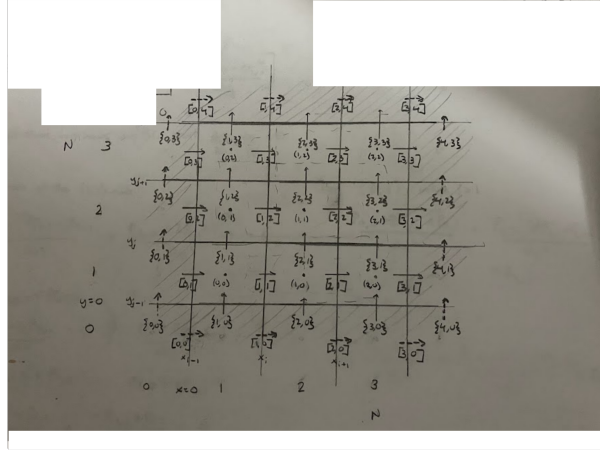


Figure 2: Illustration of staggered velocity-pressure grid.

3.2 Semi-Discrete Momentum Equation: Predictor and Corrector Step

The momentum equations, given in Eqs. 8, 13, cannot be solved in their given form as the pressure field that appears in these equations does not necessarily satisfy continuity. Accordingly, the momentum equation is split into a predictor step equation, Eq. 17, and a corrector step equation, Eq. 18, by discretizing the equations temporally. Note that “ k ” is the time-step index. An intermediate velocity field, u_i^* , is solved for in Eq. 17 which does not satisfy mass continuity. The pressure Poisson equation, 11, is then applied to this velocity field, u_i^* , as shown in Eq. 19, such that a pressure field, p , is solved for which does satisfy continuity. The corrector equation, 18, and a velocity field at time: $k + 1$ is obtained which does satisfy mass continuity.

$$\text{St} \frac{u_i^* - u_i^k}{\Delta t} = - \frac{\partial(u_j u_i)}{\partial x_j} + \frac{1}{\text{Re}} \frac{\partial^2 u_i}{\partial x_j^2} \quad (17)$$

$$\text{St} \frac{u_i^{k+1} - u_i^*}{\Delta t} = - \frac{\partial p}{\partial x_i} \quad (18)$$

$$\frac{\partial}{\partial x_i} \left(\frac{\partial(u_j^* u_i^*)}{\partial x_j} \right) = - \frac{\partial}{\partial x_i} \left(\frac{\partial p}{\partial x_i} \right) \quad (19)$$

References

1. Ferziger, J. H. & Perić, M. *Computational Methods for Fluid Dynamics* 3rd ed. (Springer, Heidelberg, 2002).